

Speak in Public: an Innovative Tool for the Treatment of Stuttering through Virtual Reality, Biosensors, and Speech Emotion Recognition

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ABSTRACT

Stuttering is a common speech disorder that affects over 80 million people worldwide. Exposure therapy (ET) is a well-established approach to treating stuttering, but it can be complex and risky to simulate challenging real-life situations. Virtual reality (VR) is a promising tool that can help overcome these limitations. This article presents Speak in Public, a system that combines VR, biosensors, and speech emotion recognition to provide objective measures of patients' stress and emotional state during ET sessions. In a preliminary study with five participants, we found that the use of VR successfully replicated stressful situations as indicated by biosensors. Additionally, speech emotion recognition showed that patients primarily experienced fear during these activities. Findings are preliminary, and further research with larger samples is needed to validate them. This study highlights the potential of VR and biosensors to enhance ET for stuttering and provides insights for future research in this area.

CCS CONCEPTS

• **Human-centered computing** → *Virtual reality*; • **Computing methodologies** → *Machine learning algorithms*.

KEYWORDS

stuttering, virtual reality, biosensors, exposure therapy, speech emotion recognition

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1 INTRODUCTION

Stuttering is a speech disorder in which the normal rhythmic flow of speech is disrupted by frequent pauses, blocks, hesitations, and repetitions of syllables, words, and sounds [29]. Worldwide, over 80 million people stutter. The challenges faced by individuals who stutter in communication and social interaction can lead to isolation and a vicious cycle of worsening symptoms [28]. In this regard, it is not surprising that 46% of people who stutter (PWS) are estimated to meet diagnostic criteria of a social anxiety disorder (SAD) as opposed to 4% of fluent speakers [3].

In treating anxiety disorders, a common approach is Exposure therapy (ET) [31], defined as “*psychotherapy that involves repeated real, visualized, or simulated exposure to or confrontation with a feared situation or object or a traumatic event or memory*”. Experiencing feared situations is considered essential to activating mental representations to promote new learning and challenge unhelpful thoughts [15]. Additionally, The Emotional Processing Theory [20] suggests that repeated exposure to feared stimuli leads to habituation and lower anxiety response. This theory is often used in treatments for stuttering, where patients are exposed to controlled verbalization situations with one or two interlocutors. However, it is difficult to replicate real-life scenarios outside of therapy to measure emotional activation. Consequentially, for PWS, the leap from clinical to real-life situations can be extremely overwhelming [19]. To overcome these issues and facilitate treatment generalization, clinicians are advised to vary the settings for practicing techniques, the people that PWS interact with, and the tasks they perform [18, 19]. However, conducting therapy sessions outside the clinic faces three main challenges: (i) the clinicians might be unable to control the external environment: PWS may experience social interactions that undermine confidence in their ability to use newly acquired techniques resulting in the risk of negatively impact both the treatment and the therapeutic relationship [6], (ii) it is often very expensive to take patients outside the clinic to practice newly learned skills or to establish complex interactions within the clinic, for instance, assembling an audience of unfamiliar people for each client who needs to practice speaking in public quickly becomes unfeasible [7], (iii) confidentiality for clients is likely to be compromised.

The use of Virtual Reality (VR) can help to fill the gaps that are present in the current ET technique by simulating experiences that are extremely close to the real world. The use of Virtual Reality Environments (VREs) can provide four main advantages to ET: (i) since exposure therapy requires repeated experience of fear-inducing situations, the ability of VR to create on-demand experiences makes it an appealing and valuable medium that easily replicates real-life scenarios and not expensive; (ii) unlike real audiences, VR offers an opportunity to present unique stimuli to users without the risk of aversive outcomes at inappropriate times; (iii) VRE is customizable: every PWS can train under the circumstances that are more challenging for them, (iv) VRE mitigates the risk of providing a treatment that is not fully effective since the communication style of therapist can be influenced by the amount of stuttering produced by the participant.

Another problem that emerges during therapy with people who stutter is their limited objectivity in providing feedback to the therapist due to a lack of self-confidence and a strong sense of anxiety about giving 'wrong' answers or starting stuttering. This problem can be tackled through the use of biosensors. These devices are widely used in therapy to keep track of a patient's biological progress by collecting data such as body temperature, heart rate, and Electrodermal Activity (EDA). In specific, the latter refers to the variation of the electrical conductance of the skin in response to sweat secretion also called skin conductance response (SCR). Change in SCR is produced by our sympathetic nervous systems which drive aspects of human behavior, as well as cognitive and emotional states. It's acknowledged that there is a strong association between the EDA signal and emotional arousal (a state of heightened physiological activity) [1, 17]. Thus, properly combined and processed data can provide powerful insights and information about the patient's stress state, making it much easier for the therapist to see which stimuli are the most 'triggering' for the PWS.

On the other hand, to the best of our knowledge, speech emotion recognition has never been integrated with VRE, but it can offer valuable information about which scenarios are more frightening for each patient and provide insights into the comorbidity between stuttering and social anxiety.

This article presents *Speak in Public*, a system integrating virtual reality, biosensors, and speech emotion recognition in a unique way by providing therapists with objective measures of the patient's stress and emotional state during stuttering therapeutic sessions.

2 STATE OF THE ART

During the past few decades, there has been an increasing amount of research about the use of VR for ET to overcome specific phobias [4, 23] and anxiety [2, 5, 22, 30]. However, there is still much to explore about the effectiveness of VREs for stutterers. To date, only five articles have been presented on this innovative modality: four with adults [8–10, 29] and one with children/adolescents [24]. [9] was the first study to investigate how VREs can be useful for stuttering therapy. In that research, 23 PWS were exposed to job interview scenarios with and without VR. One of the most interesting results obtained indicates that the interviewer's communication style is affected by the amount of stuttering produced by participants, making the therapy less effective with time. Additionally, participants'

subjective reports of the VR experience indicate similar reactions to those they report experiencing in the real world. Further, [8] asked 10 adults, who stutter, to give a talk in front of a virtual audience and in front of a virtually empty room. The participants reported higher subjective levels of anxiety when speaking to the virtual audience compared to the empty room. This result proves that VREs have utility in elevating subjective ratings of distress in PWS by simulating objectively everyday life stressful situations. This result was confirmed by [29] which developed a Virtual Reality Exposure Therapy (VRET) tool that slowly introduced the user into a lecture theatre environment where they were to give a talk to an animated audience on a particular topic. The results gathered throughout the VRET regarding anxiety level, body temperature, and electrodermal activity (EDA) showed the participants becoming anxious within the VRET and their anxiety levels decreasing over repeated sessions. To the best of our knowledge, this is the only study up to date that merges the use of VRE with biosensors for PWS; however, it only analyses EDA's raw data. Regarding speech emotion recognition, it has already been successfully associated with virtual reality environments [25], but no examples have been found that explore this combination for stuttering treatments. Our methodology brings together virtual reality, biosensors, and speech emotion recognition at the scope of supporting VRET.

3 SYSTEM OVERVIEW

The *Speak in Public* system is composed of 4 components: (i) a Virtual Reality application, (ii) an Android application that allows data collection from the biosensor, (iii) a set of algorithms for biophysical signal analysis and emotion speech recognition, and finally (iv) the dashboard that allows the therapist to control the execution and post-session evaluation.

3.1 VR application

The VR application runs on MetaQuest 2 devices and was designed together with the therapists using an iterative co-design process where the questions to be used as stimuli and the training environments were decided upon. The activities are realistic simulations that recreate Q&A situations in which the user is assigned a certain amount of time to talk about a given topic. These scenarios offer the user a VRE in which they can train their public speaking skills in a safe and controlled way. Currently, the *Speak in Public* VR application contains five scenarios: bar, party with friends, doctor's waiting room, classroom, and job interview. While in the experience, the user receives new topics or questions to answer within the app in two formats: text-based or speech-based. Question generation can be random or left to the therapist who can follow the whole user experience in real time from the web platform. The user's answers are only verbal and are recorded by the platform for future analysis (i.e., stuttering activity detection and speech emotion recognition). Each experience can be further customized by the therapist who can choose to add or remove distractors. There are two categories of distractors: visual (lights that go out, arrival of the waiter to distract the user) and audio (yawning, a door slamming, a phone ringing). The therapist knows when to activate a distractor as he can see what the user is experiencing in virtual reality. At the end of

the session, the whole VR experience is saved (audio and video) on the platform, and the therapist can recheck it in a second moment.

3.2 Biosensor and Android application

The Android application was designed ad-hoc for the biosensor used in this study (Empatica E4). The purpose of the app is to connect to the biosensor and enable the display and storage of data during a Speak in Public session both on the device where the app is installed and on the therapist's web platform. Empatica was chosen among several biosensors because it is the only wearable on the market to combine EDA and PPG sensors, simultaneously enabling the measurement of sympathetic nervous system activity and heart rate. In addition to these data, the biosensor is equipped with a 3-DOF accelerometer, an infrared thermopile for reading skin temperature, and an internal real-time clock.

3.3 Therapist Dashboard

The dashboard allows therapists to start new sessions and review past ones. Once a session has been started, the therapist can monitor the user experience in the virtual environment in real-time, being able to activate and deactivate distractors and questions to stimulate the user. During the session, the therapist can also visualize data from the biosensor. Since raw data from the speech and the biosensor is not easy to read and interpret on the fly, the dashboard offers the therapist a set of tools in the post-session phase to infer the state of stress experienced by the patient. Once a session is over, in fact, the therapist can: (i) download the audio and the biological data collected, (ii) re-watch the entire session, (iii) make further analysis of the gathered data by applying the algorithms described in Section 3.4, and visualize the results on graphs.

3.4 Algorithms

To obtain more readable and meaningful insights for the therapist a set of data analysis algorithms was developed. Indeed, simple raw data from the biosensor or the audio recorded during the activity may be of low utility in this form. For this reason, three algorithms have been implemented: (i) the first to decompose the EDA signal into its main components, (ii) the second to identify the patient's state of arousal during the session and to infer the user's state of stress/not stress (iii) the third to recognize emotions from the tone of voice.

The EDA signal results from two additive processes: a tonic base level driver, which fluctuates very slowly (seconds to minutes), and a faster-varying phasic component (fluctuating within seconds). The first algorithm derives these two components from the EDA signal. When there are variations in phasic activity, they occur in the form of peaks. The EDA peaks reflect the greater emotional arousal experienced during a specific event. It is important to note that while more EDA peaks indicate greater emotional intensity, the direction of this emotion (valence) can not be derived from EDA alone. The EDA signal is, therefore, not representative of the type of emotion but simply the intensity of it. In order to better understand the emotion that generated the peak, a second algorithm was developed. This consists of a machine learning model capable of differentiating between stressful and non-stressful conditions starting from segments of biological signals acquired through

Empatica's E4. The starting point of the model was the work reported in [27]: an emotion recognition benchmark trained on the WESAD dataset using well-known features and standard machine learning models [26]. The choice of a specific biosensor - Empatica E4 - implies the need of using a dataset that either contains data acquired by this same device or has at least been validated for its use. WESAD (Wearable Stress and Affect Detection) [27] has the great advantage of containing physiological signals measured through the device chosen for our project, the E4. Indeed, the fact that the sensor used at inference time is the same as the one used at training time is certainly an aspect that will benefit the performance of our algorithm [21]. In addition, WESAD is the only publicly available dataset containing both stress, considered as an emotion, and emotion data. More specifically, each subject experienced three conditions: baseline (neutral reading task), amusement (watching a set of funny video clips), and stress (public speaking). In our case, the specific task required us to train the Machine Learning model by dividing the WESAD dataset into two labels (1: stressed and 0: not stressed). Where condition 0 (not stressed) included both the baseline and amusement conditions. The algorithm is thus trained to recognize a stressed situation as opposed to a neutral or happy one. To tune the model parameters, not having a particularly large dataset, we chose to use cross-validation. Indeed, this technique is known for its "efficient" use of data, as every observation is used for both training and testing. More specifically, to ensure that we choose a model with good generalization capabilities on unseen subjects, we adopted the leave-one-subject-out cross-validation (LOSO-CV) scheme. Using a Neural Network classifier, we obtained an 89.2% accuracy on the bi-affective classification problem (stress vs. amusement & neutral), with a 30-second output temporal resolution. Having used leave-one-subject-out cross-validation, our results suggest that generalization to unseen subjects is possible. The implemented training algorithm follows the traditional steps of any classification system: pre-processing, signals segmentation, feature extraction, and classification.

3.5 Speech Emotion Recognition

To extract emotions from speech, we used an original model that we developed ad-hoc through an iterative process [12, 13]. The model takes an audio file as input, extracts a contextualized representation of it based on the wav2vec 2.0 model [16], and returns a vector of probabilities associated with the considered emotions as output (i.e., joy, fear, anger, sadness, disgust, neutral, and surprise). We obtained a pre-trained wav2vec 2.0 model checkpoint from Hugging Face ¹, which we fine-tuned on Italian using the Common Voice dataset², resulting in a final word error rate (WER) of 9.36%. Since wav2vec 2.0 cannot output a sentence representation [16], we had to fine-tune it for utterance-level classification tasks. We evaluated multiple architectures for the final layer of the model and eventually chose one that was simple to process the contextualized representation extracted by wav2vec 2.0. This included a global average pooling across the time dimension (to manage audio recordings of varying lengths), ReLU activation, and a single linear layer to predict the emotion categories as in [14]. We fine-tuned the model for speech

¹<https://huggingface.co/jonatasgrosman/wav2vec2-large-xlsr-53-italian>

²<https://commonvoice.mozilla.org/it/datasets>

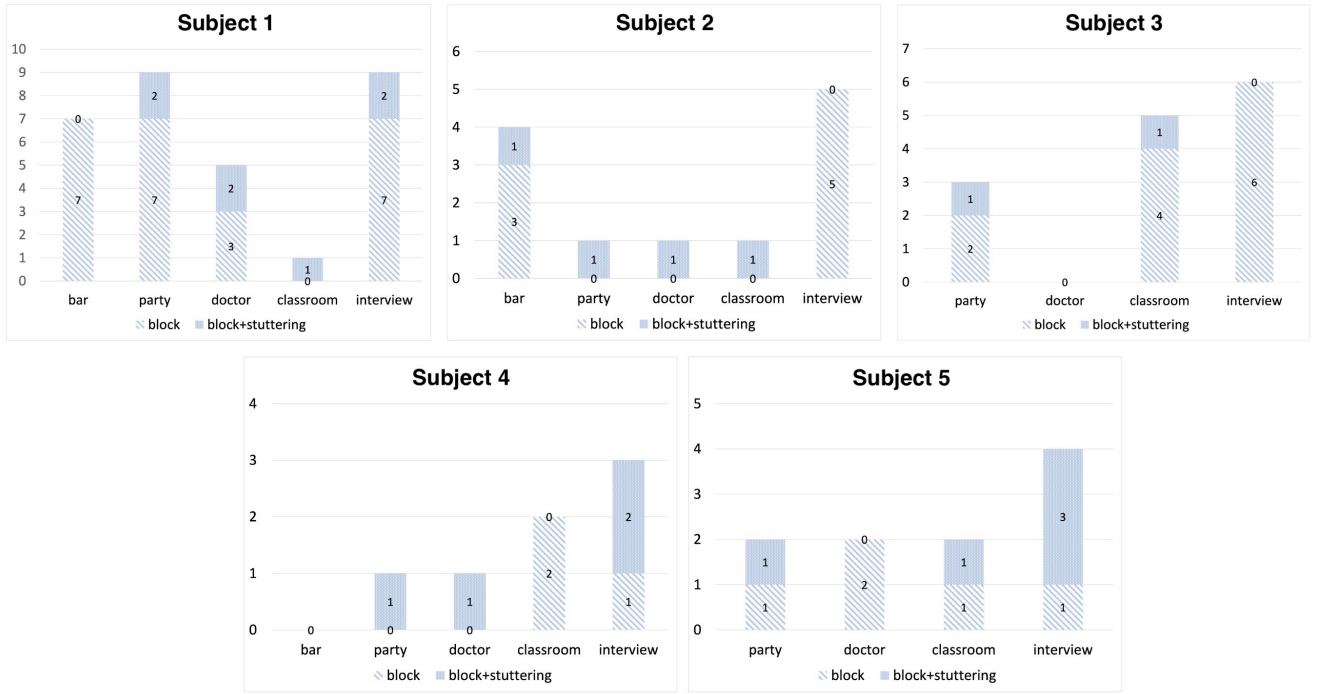


Figure 1: Overview of the peaks of EDA phasic concurrently with stuttering events. The X-axes represent the different scenarios (bar, party, doctor, classroom, interview), and the Y-axes represent the number of EDA peaks experienced.

emotion recognition on the largest Italian emotional speech corpus [11], and tested it on a portion of the same dataset not used for training. The overall accuracy of the model reached 83%.

4 EXPLORATORY STUDY

We ran an exploratory study that investigated which insights can emerge from the measures extracted from biosensors and speech during VR-based exposure therapy and how such information can benefit the treatment of stuttering.

4.1 Participants

The sample comprises 5 males who stutter (mean: 18, sd: 2.097) and underwent VR training, recruited voluntarily from the CRC Balbuzie in Rome. The selection was made according to the following inclusion criteria: (i) diagnosis of stuttering (mild to moderate), (ii) age: 15-19 years, (iii) Italian mother tongue, (iv) absence of anxiety disorders. All subjects were preliminary assessed through the administration of the SSI-4 test (Stuttering Severity Instrument-4th edition) for the evaluation of the severity of stuttering and the RCMAS-2 questionnaire (Revised Children's Manifest Anxiety Scale- 2nd edition) for the exclusion of an anxiety disorder.

4.2 Procedure

The study took place at CRC Balbuzie in Rome, in rooms equipped with therapeutic materials, desks, and chairs. Before the training, all recruited subjects were exposed to a first practice phase in which they learned to use VR and were evaluated on the possible effects

of Motion Sickness. Subsequently, a stuttering assessment was performed in a pre-test phase. All our patients underwent five sessions with Speak in Public during which they experienced different scenarios: bar, party, doctor, classroom, and interview. The order - from the most to the less comfortable - was the same for all testers, and was defined by the therapists. During each session, the therapists were in control of the VR stimuli and chose sentences and questions from the interactive panel of the application, according to the protocol defined during the study design. They recorded the occurrence of the different types of disfluencies (block and stuttering) manually. Physiological measures and voice recordings were collected automatically by our system.

5 RESULTS AND DISCUSSION

5.1 EDA peaks and disfluencies

We used the algorithm described in Section 3 to identify the peaks of EDA that correspond to moments of maximum stress, and then, we analyzed if these peaks were in concomitance with a stuttering event. We consider two macro-categories of stuttering events: "block" (a stoppage of speech, when someone gets suddenly "caught" on a particular (part of) a word that makes it difficult to continue speaking) and "block+stuttering" (repetitions of syllables, words, or sounds, typically anticipated by a "block"). We detected many moments of stress, which confirms the results obtained by previous studies (e.g., [8]) indicating that VR can be used to simulate stressful and challenging situations (and train people to manage them). The number of moments of stress (indicated on the Y-axis of the

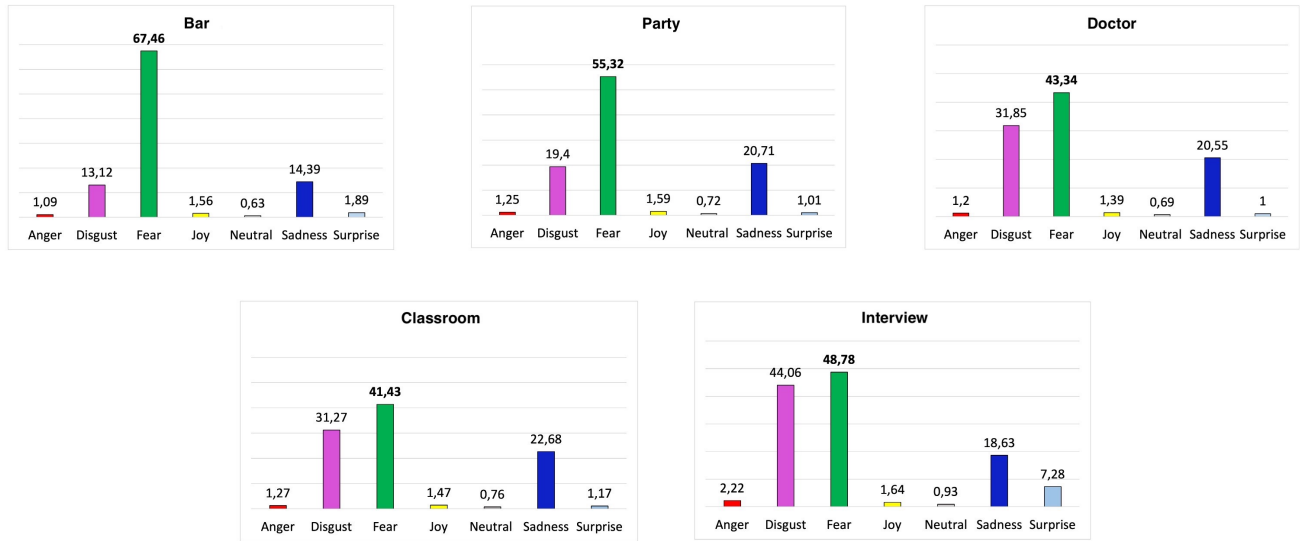


Figure 2: An aggregated overview of the emotions extracted from speech during each task. The X-axis represents the emotions measured from speech (anger, disgust, fear, joy, neutrality, sadness, or surprise), and the Y-axis represents the probability of each of them.

diagrams in Fig. 1) differed among the different scenarios (indicated on X-axis). The job interview scenario was the most stressful for all participants: during this VR experience, users were asked direct questions about their stuttering status (e.g., "Why are you stuttering so much?") which - as expected - created some discomfort. We also analyzed the temporal correspondence between stuttering events and stress. Our data suggest that all stuttering events occurred when the user was in a stressed state while speaking. Figure 1 presents the number of stuttering events of the 2 different types for each user and each scenario.

5.2 Emotional analysis from speech

Our speech emotion recognizer extracted the core emotions (anger, disgust, fear, joy, neutral, sadness, surprise) experienced by each participant during each scenario (bar, party, doctor appointment, classroom, and interview) in the VR sessions. Figure 2 shows the probability that each emotion manifested per scenario aggregated by participant. The plots highlight that fear was generally the most dominant emotion, followed by disgust and sadness. These results may suggest that the experience in Speak in Public can support exposure therapy by stimulating a variety of negative feelings that patients could learn to master in the safe and controlled environment provided by VR.

Figure 3 offers a finer overview of how each participant experienced each of the core emotions during the five scenarios. We can observe that the most relevant emotions that participants experience are consistent across each scenario. Subjects 1 and 2 mainly experience one negative emotion regardless of the scenario, which is disgust and fear, respectively. The other participants mainly experienced negative emotions as well. However, we can notice that the level of sadness, disgust, and anger experienced are closely related to the scenario. This result shows that speech emotion recognition

analysis has the potential to capture the uniqueness and subjectivity of each patient, which is an important step toward personalized intervention.

6 CONCLUSION

Our exploratory study has opened a new chapter for stuttering therapy, with potential applications in tracking exposure therapy progress, understanding comorbidity with social anxiety, and empowering users with emotional self-awareness. Our promising results pave the way for future research in the field of stuttering therapy and highlight the importance of integrating technology with evidence-based treatments. In particular, we successfully integrated VR, biometric data analysis, and speech-emotion recognition to support stuttering exposure therapy. We demonstrated that VR could effectively simulate challenging scenarios that trigger stress in participants. By tracking EDA peaks, we were able to quantify stress levels experienced by participants, which further emphasizes the potential role of our methodology in stuttering treatment. The use of speech emotion recognition also provided insight into how people who stutter feel when exposed to different activities. The present study is to be considered exploratory due to the limited sample size of only five participants and the absence of a control group. Hence, the results obtained from this study must be interpreted with caution and cannot be generalized to a larger population. To overcome these limitations, we are currently planning a follow-up study that will involve a larger sample size and incorporate a control group to provide a more rigorous evaluation of the effectiveness of the proposed methodology. By addressing these limitations, we aim to validate our findings further and provide more robust evidence to support the potential use of VR, biometric data analysis, and speech-emotion recognition in stuttering therapy.

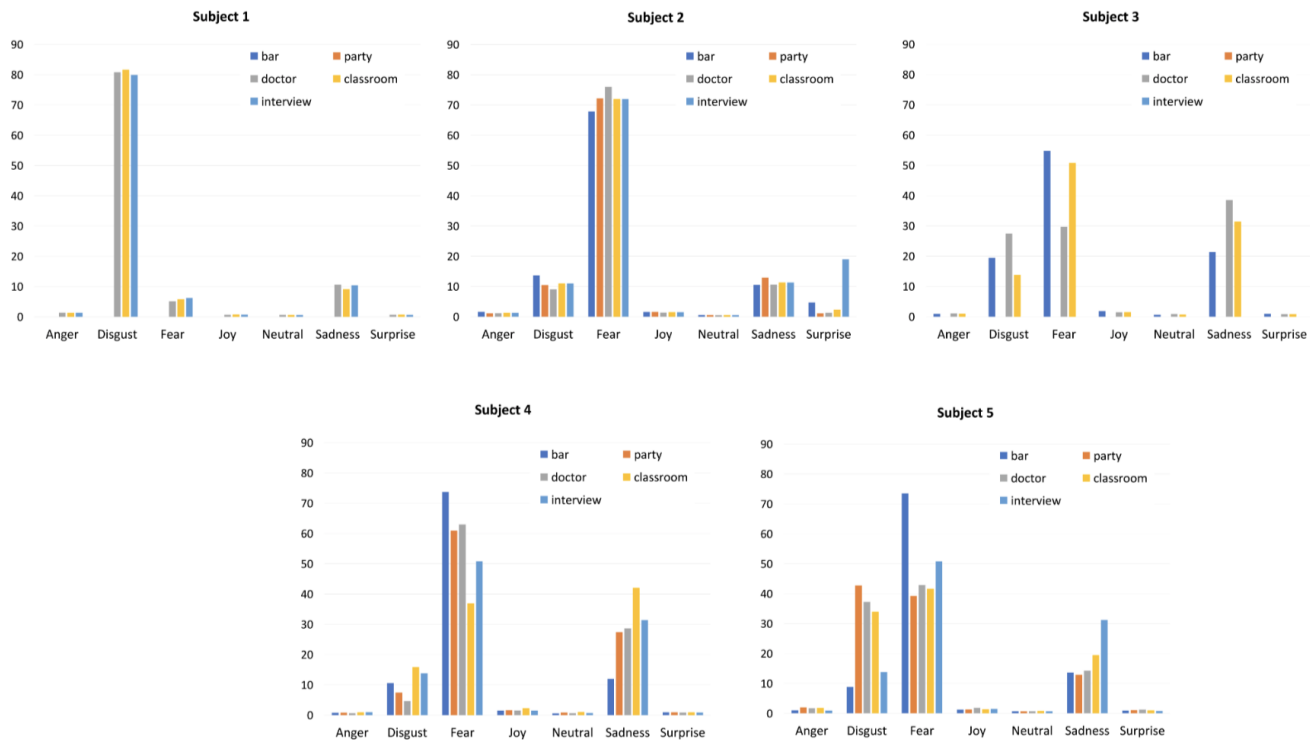


Figure 3: An overview of the emotions recognized for each participant and each scenario. The X-axis represents the emotions measured from speech (anger, disgust, fear, joy, neutrality, sadness, or surprise), and the Y-axis represents the probability of each of them. We excluded from the analysis the bar and party scenarios for Subject 1 and party and interview scenarios for Subject 3 due to low audio quality.

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